

MAINT-AI Synthetic Troubleshooting Guide

Centrifugal Pump — CP-200

Document Notice

Synthetic demonstration document created for the MAINT-AI Sprint 2 RAG prototype. It is not a manufacturer-issued service manual.

1. Equipment Overview

Equipment: Centrifugal Process Pump

Model: CP-200

Typical application: transfer of process water and compatible fluids

Normal operating status: Operational

2. Routine Maintenance

Inspect the pump and piping for leakage, monitor vibration and bearing temperature, check coupling alignment, and inspect the suction strainer according to the maintenance schedule. Confirm that the pump is adequately supplied with fluid before operation.

3. Fault: PUMP-LOW-03 — Low Discharge Pressure

Symptoms: discharge pressure is below the expected operating range and flow may be reduced.

Possible causes include a blocked suction strainer, insufficient suction supply, incorrect valve position, air entering the suction line, worn impeller components, or incorrect pump rotation.

4. Troubleshooting: PUMP-LOW-03

1. Check the suction and discharge valve positions.
2. Inspect the suction strainer for blockage.
3. Verify that adequate fluid is available at the suction source.
4. Check the suction line for air leaks.
5. Verify pump rotation after electrical isolation and according to site procedures.
6. If the impeller is suspected to be worn or damaged, stop the pump and escalate for inspection.

5. Fault: PUMP-VIB-04 — Excessive Vibration

Possible causes include coupling misalignment, cavitation, bearing wear, loose mounting, or an unbalanced rotating assembly. Check operating conditions and mounting hardware. Persistent vibration should be investigated by qualified maintenance personnel.

6. Safety

Never work on a pressurized pump or piping system. Isolate electrical power and relieve pressure before maintenance. Do not operate the pump without adequate fluid supply when dry running could damage the equipment.